

Software design and testing with C#

Formative Assessment 2 2026

Belinda Mokena

20240611 SDT621

Executive Summary

The University Learner Management System (ULMS) underwent functional testing focusing on Login and Marks Capture modules. Testing identified key defects impacting calculation accuracy, input validation, and security. Corrections were implemented and verified. System is ready for deployment pending final sign-off.

Testing Objectives: System Purpose

Testing Goals:

- Confirm all four modules work as expected with valid and invalid inputs
- Confirm the average calculation produces a consistent result and correct answer
- Verify the UI doesn't crash when users enter wrong data

Scope of testing

- **In Scope:** Student Registration, Course Enrolment, Marks Capture, Reports modules - testing only the application interface and its responses.
- **Out of Scope:** Database functionality, user authentication, multi-user access, and performance under large data volumes. Testing was limited to the desktop application UI only.

Testing Type	ULMS Example	Approach
Functional Testing	Login Button	Check that clicking Login responds without crashing the app, regardless of input
Input Validation Testing	Marks entry textbox	Enter 991, -5, abc, and leave blank to see if the app handles invalid input gracefully
Usability Testing	Button labels and layouts	Confirm all buttons are labeled clearly and the form layout is easy to navigate
Reliability Testing	Average calculation	Enter marks 70, 80, 90 three times and check if the average always shows 80

Testing Environment Setup

Hardware:

- Windows PC, 8GB RAM minimum

Software:

- IDE: Visual Studio 2022 for running the ULMS application
- OS: Windows 11 Pro 64-bit

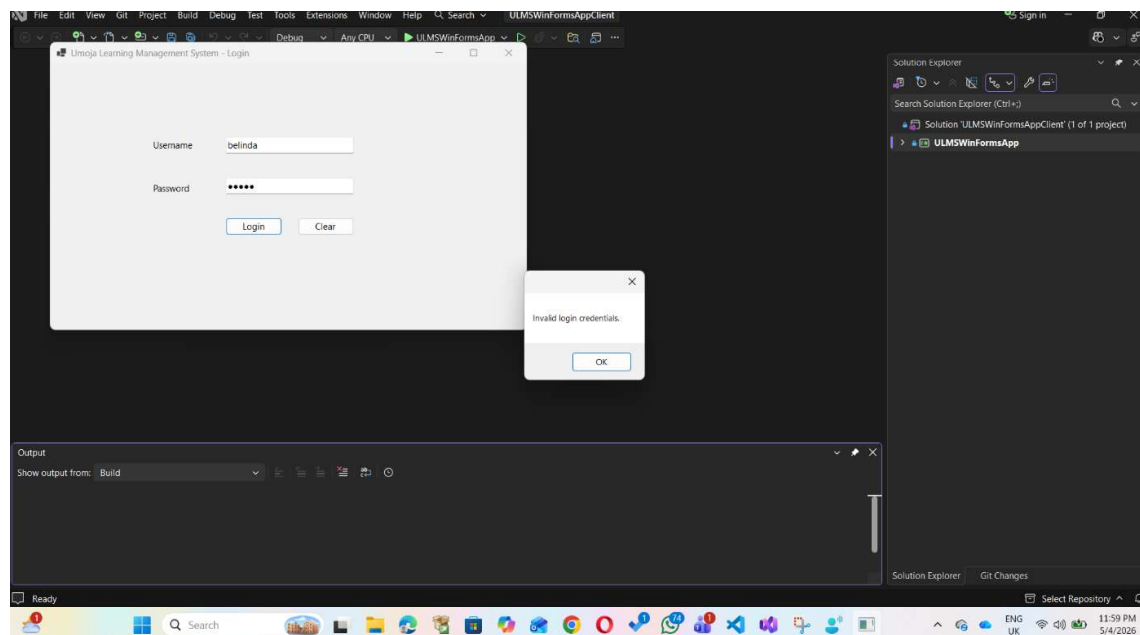
Test Data Sets:

- Valid: Marks 70, 80, 90
- Invalid: Marks 101, -5, abc,
- Boundary: Marks 0 and 100

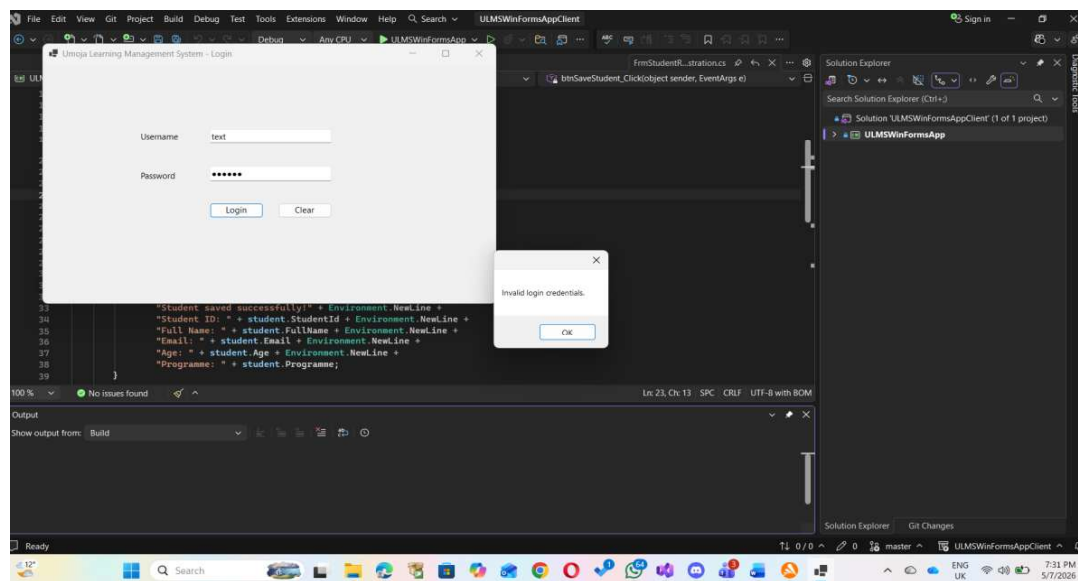
Test Case Design - Task 2

Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC01	Valid login attempt	Username: admin, Password: admin	Application opens main dashboard without error	ULMS Dashboard opened with 4 buttons: Student Registration, Course Enrolment, Marks Capture, Reports	Pass
TC02	Invalid login attempt	Username: test, Password: wrong	Application remains on login screen and does not crash	The "Invalid login credentials" popups displayed and application did not crash	Pass
TC03	Empty login fields	Username:, Password:	Application remains on login screen and does not crash	"Invalid login credentials" popup displayed. Application did not crash	Pass

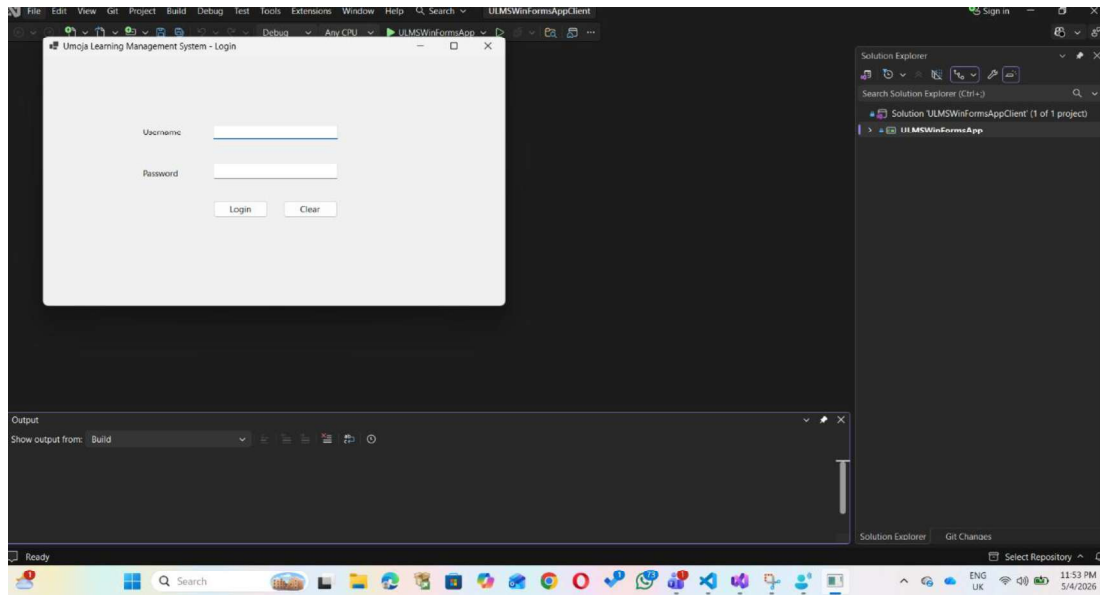
TC01



TC02



TC03

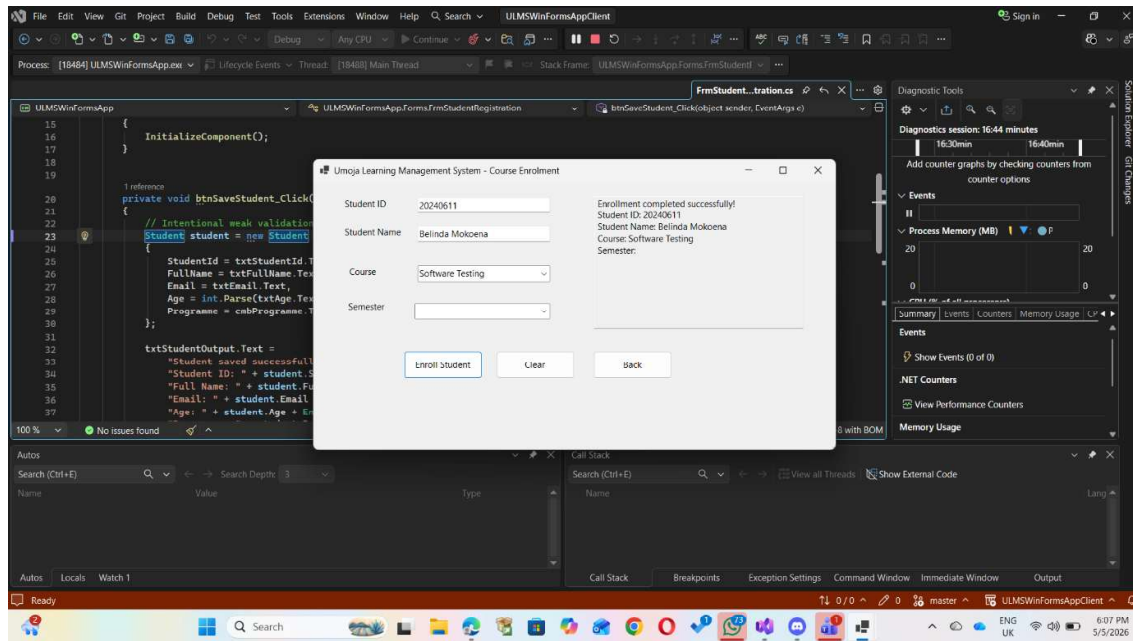


Course Enrolment

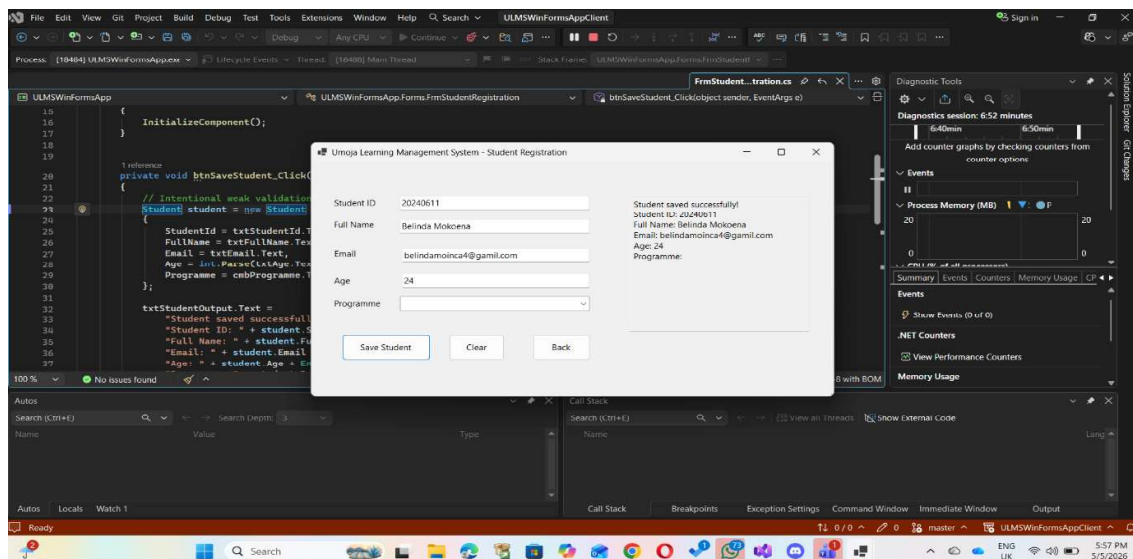
Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC04	Valid course enrollment	Student ID: 20240611, Course: Software Testing, Semester: Empty	Course is enrolled and displayed in the list	"Enrolment completed successfully!" displayed with Student ID, Name, Course. Semester blank	Fail
TC05	Valid course enrolment with semester	Student ID: 20240611, Course: , Semester: Semester 2	Course is enrolled and displayed in the list	"Enrolment completed successfully!" displayed with Student ID, Name, Semester 2. Course blank	Fail
TC06	Empty course and Empty semester	Student ID: 20240611, Course: , Semester:	Application does not crash. No action is performed	"Enrolment completed successfully!" message displayed with Student ID and Name populated. Course and	Fail

				Semester fields are blank	
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TC04



TC05

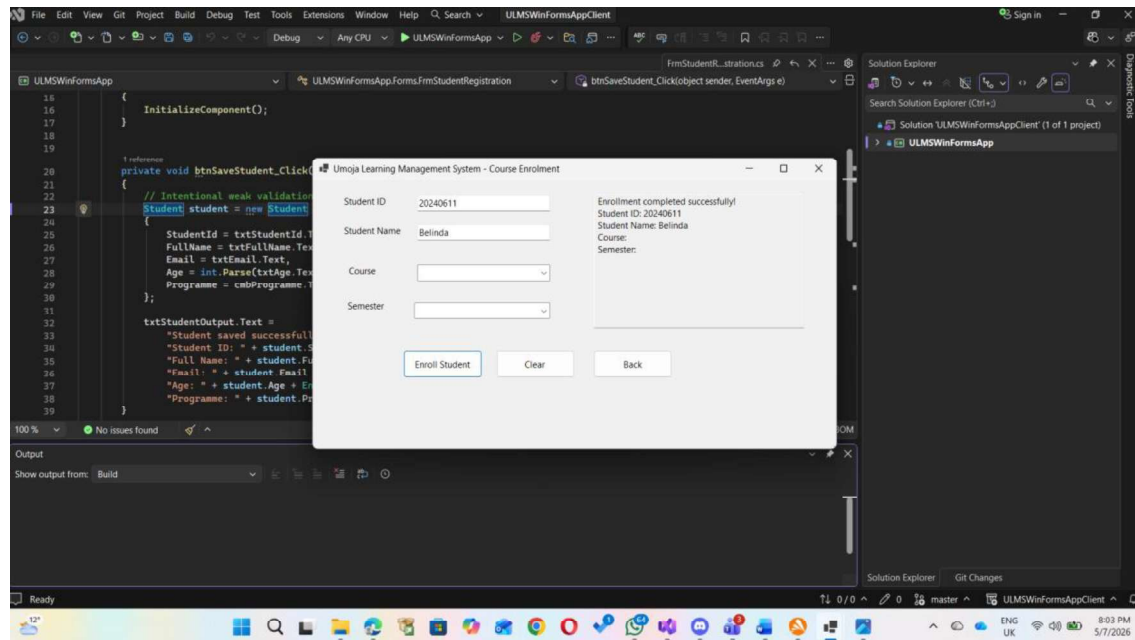


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TC06

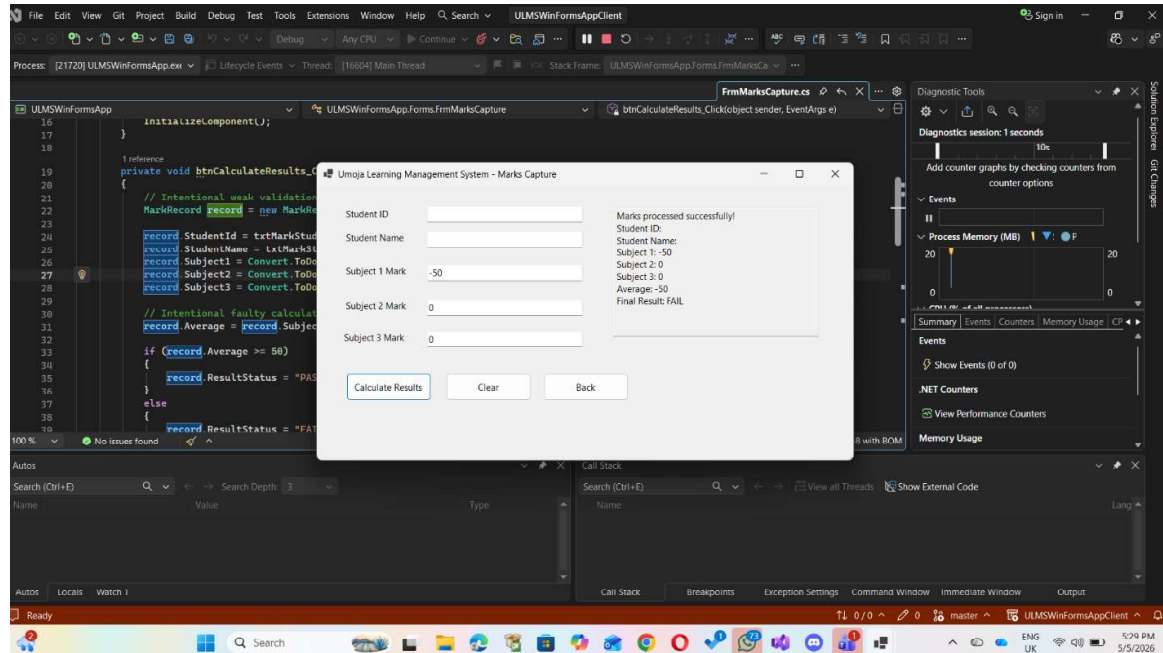


Marks Capture - Numeric Validation

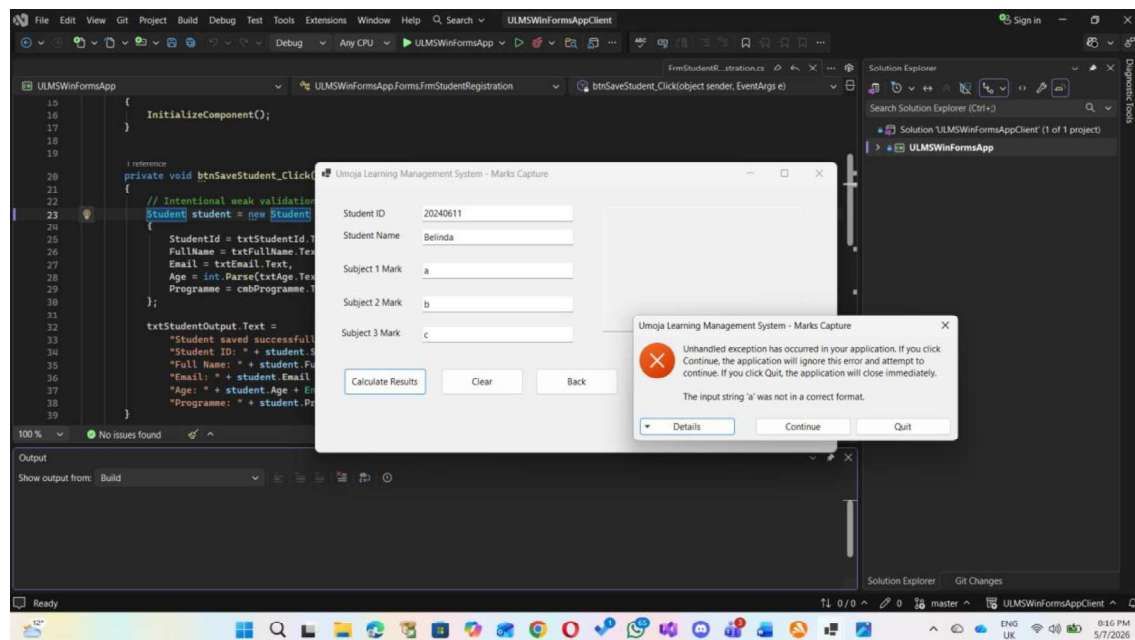
Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC07	Valid numeric marks	Marks: 50, 0, 0	Marks are accepted and average calculates correctly	Subject 1: 50, Subject 2: 0, Subject 3: 0 entered. Average: 50, Final Result: PASS	Pass
TC08	Non-numeric marks input	Marks: abc	Application does not crash. Input is rejected or ignored	Unhandled exception occurred. Error message: "The input string 'a' was not in a correct format" displayed when non-numeric marks were entered	Fail
TC09	Out of range marks	Marks: 101	Application does not crash. Input is rejected or ignored		
TC10	Negative marks input	Marks: -50, 0, 0	Application does not crash. Input is rejected or ignored	Subject 1: -50, Subject 2: 0, Subject 3: 0 accepted. Average: -50, Final Result: FAIL	Pass
TC11	Empty marks field	Marks:	Application does not crash. No	System.FormatException thrown when empty string entered	Fail

			action is performed		
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TC07



TC08

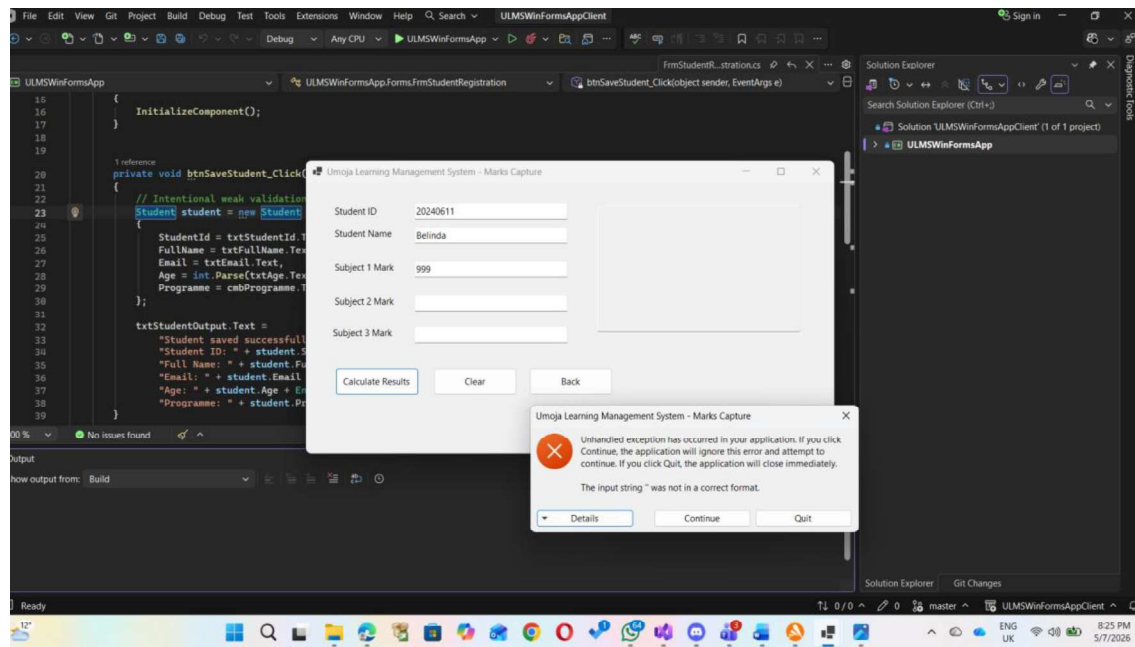


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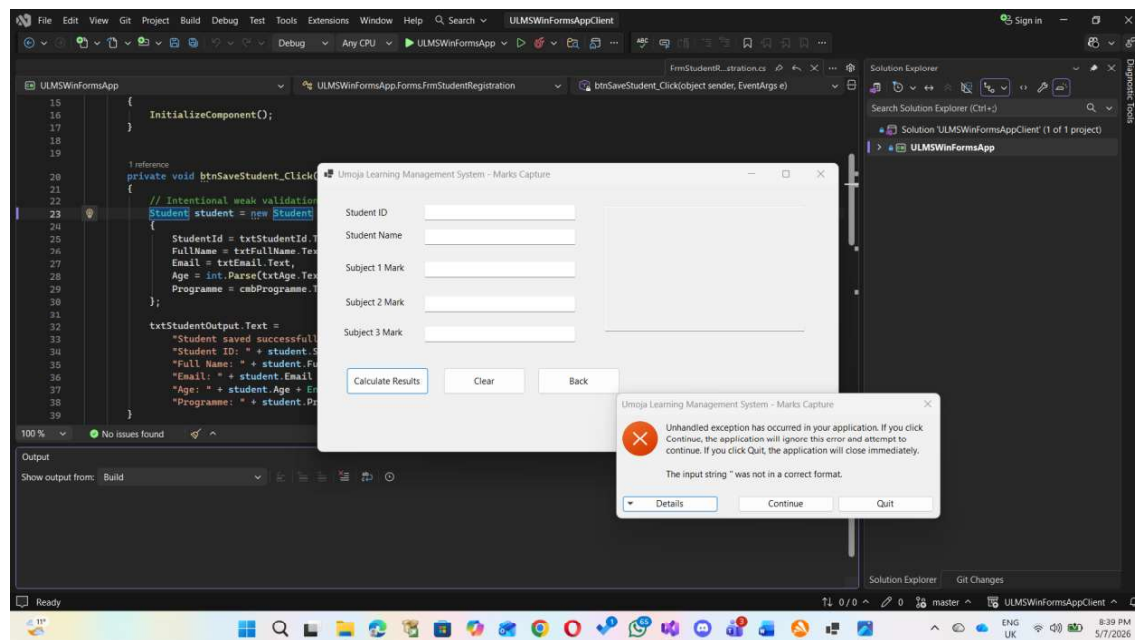
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TC10

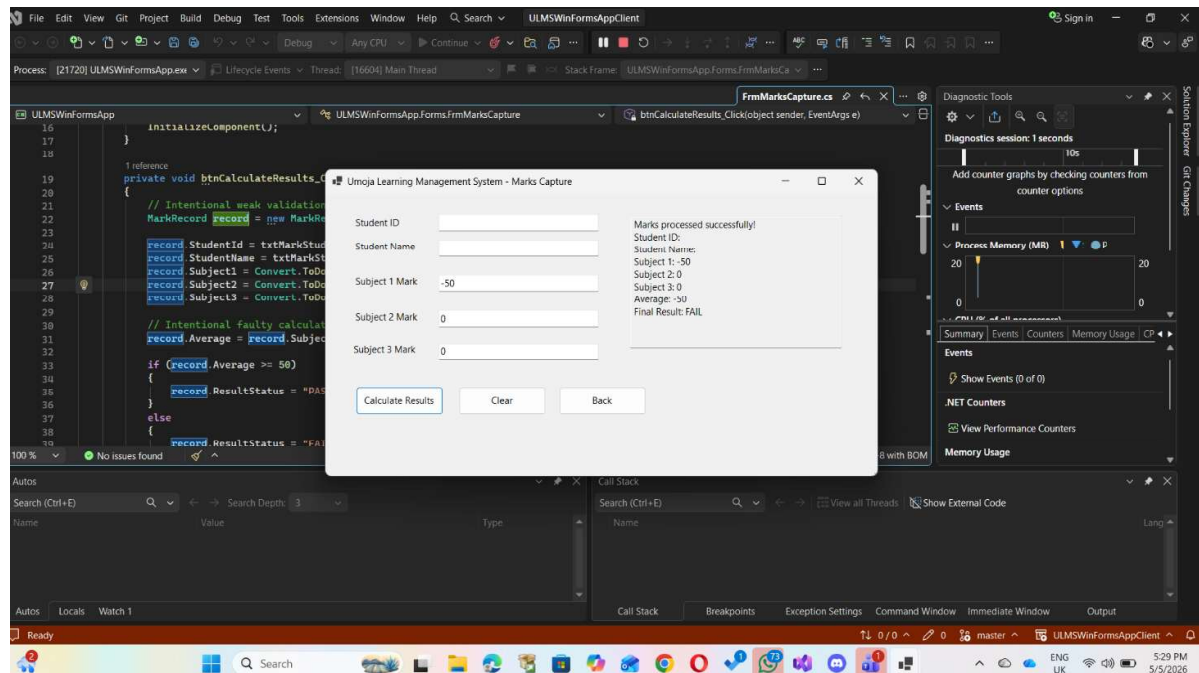


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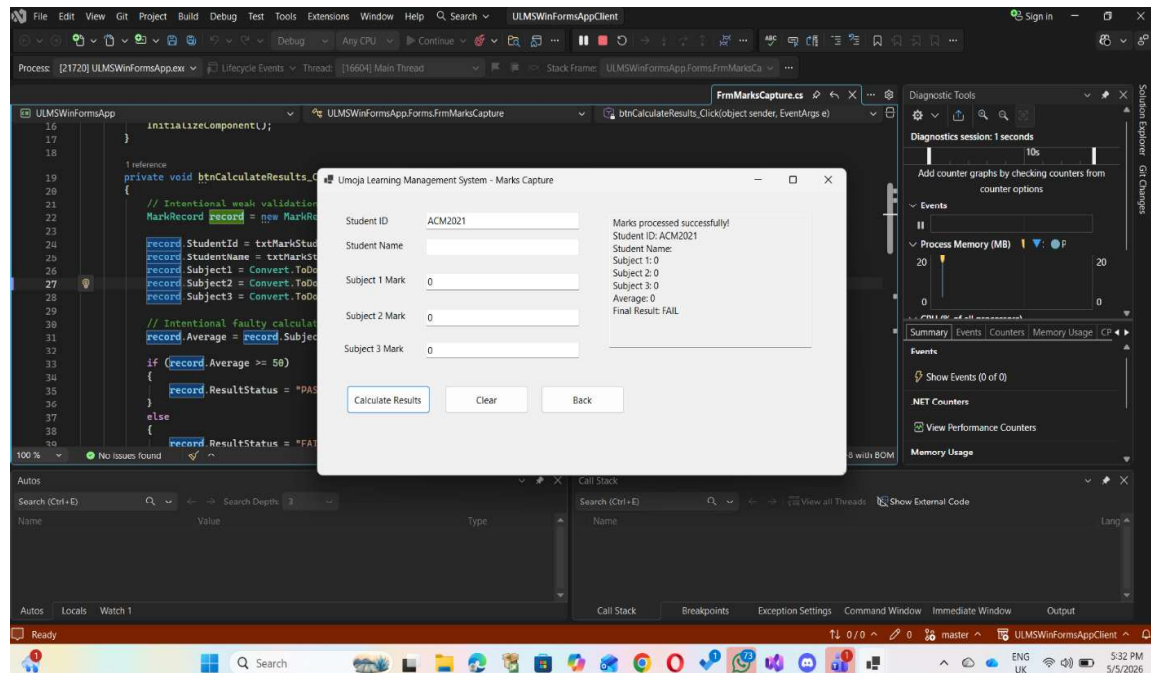
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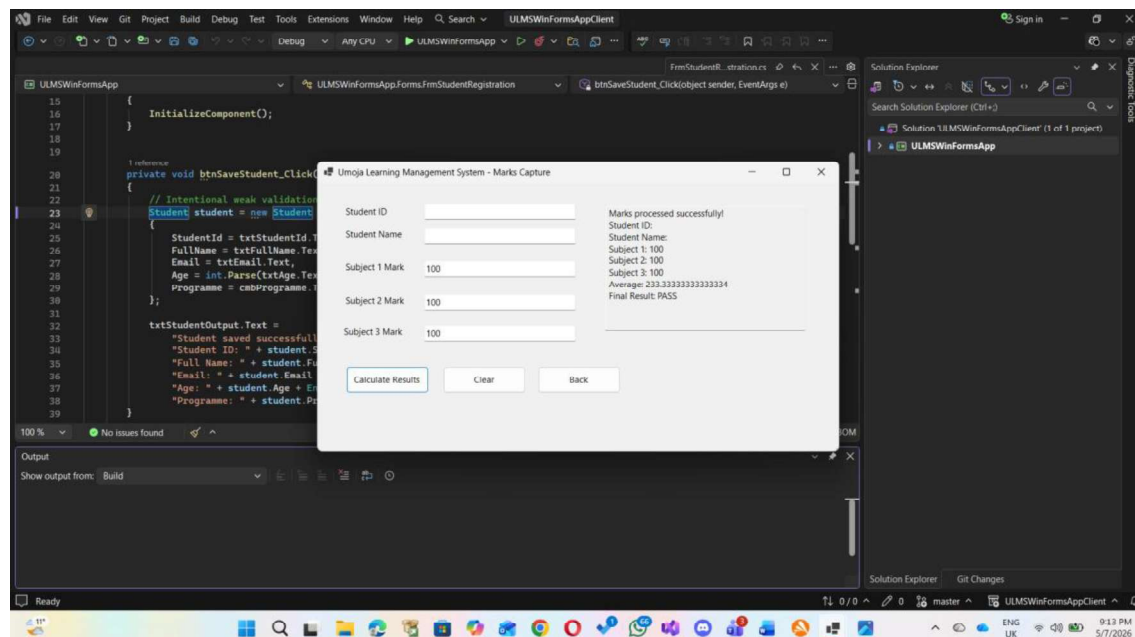
Average Calculation - Boundary Testing

Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC12	Boundary value at 0	Average displays 0.00 consistently	Average displays consistently	Student ID: ACM2021, Marks: 0, 0, 0. Average: 0, Final Result: FAIL	Pass
TC13	Boundary value at 100	Marks: 100, 100, 100	Average displays 100.00 consistently	Marks processed successfully! message displayed with Subject 1: 100, Subject 2: 100, Subject 3: 100, Average: 233.33334, Final Result: PASS	Fail
TC14	Reliability check	Marks: 90, 99, 85	Average displays 91.33 consistently	Marks: 90, 99, 85. Average: 91.33334, Final Result: PASS	Pass

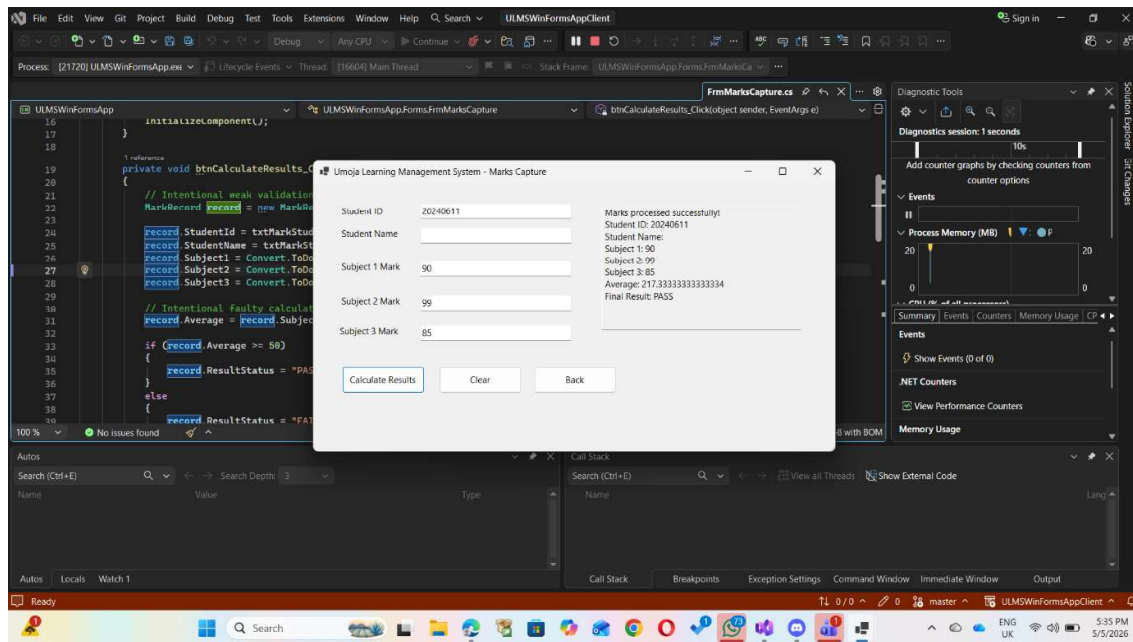
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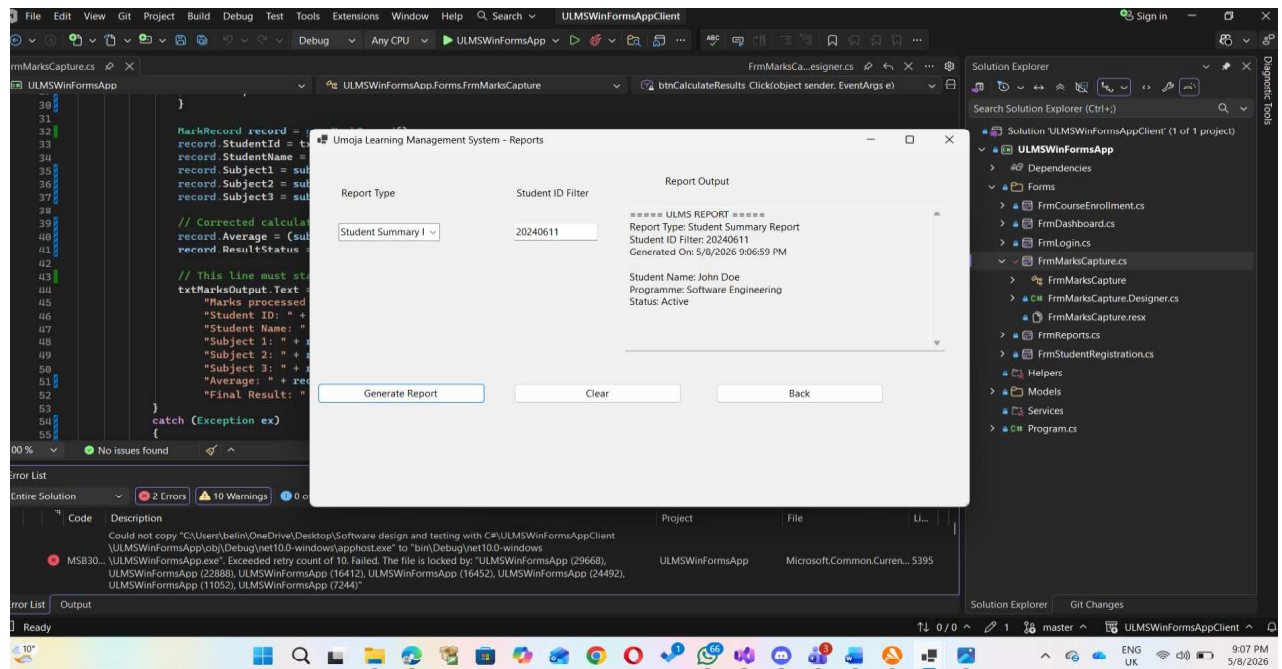


Report Generation - Output Correctness

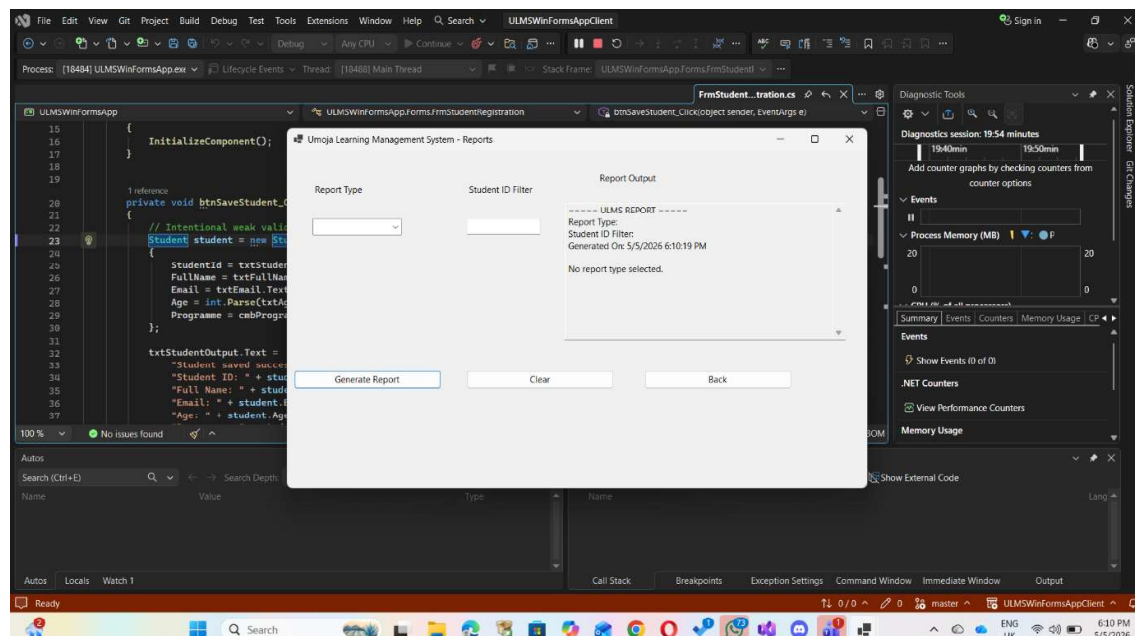
Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC15	Valid Student Summary report	Report Type: Student Summary, Student ID: 20240611	Report displays student details and status	Report generated: Student Name Belinda Mokoena, Programme Software Engineering, Status Active	Pass
TC16	Report with no report type selected	Report Type: , Student ID: 20240611	Report displays message without crashing	"No report type selected." message displayed in Report Output. Application did not crash	Pass
TC17	Valid Marks Report	Report Type: Marks:	Report displays	Report generated:	Pass

		Subject 1:78, Subject2:65, Subject3:80 Report, Student ID:	marks and average	Subject 1: 78, Subject 2: 65, Subject 3: 80, Average: 169	
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TC15



TC16

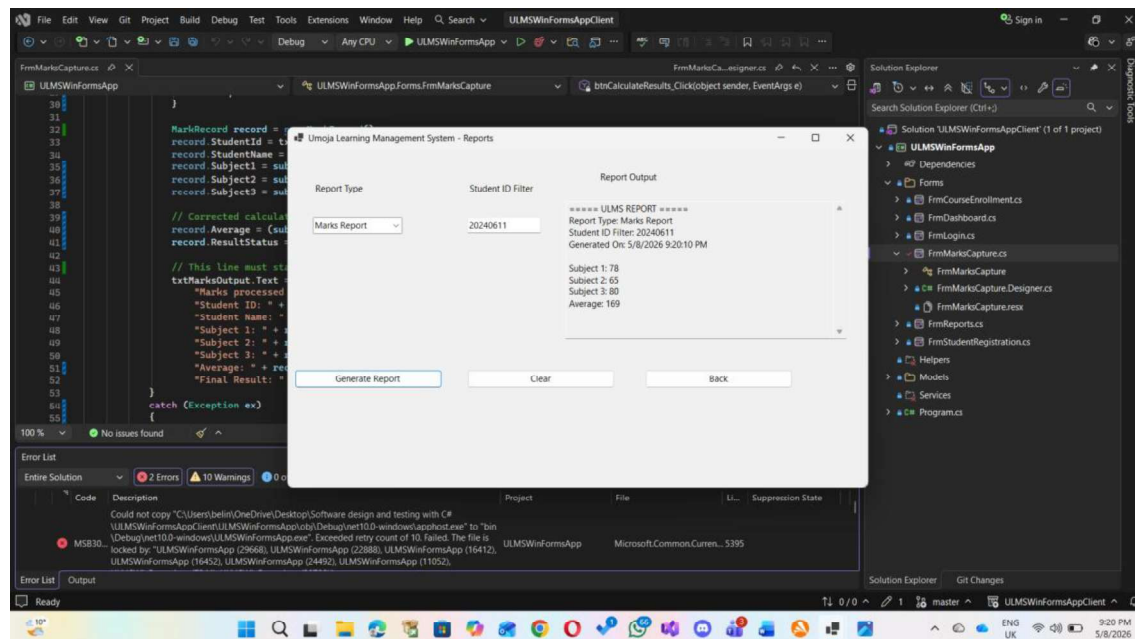


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TC17



Performance Testing - Report Loading Time Method:

Opened the ULMS app → Navigated to Reports → Clicked Generate Report → Used my phone stopwatch to time how long it took for the report to display.

Result: Report loaded in approximately 3.6 seconds on average

Interpretation: Loading time is under 7 seconds, which is acceptable for a desktop app. No performance issues observed.

Usability Testing - Navigation Steps Method:

Counted the number of clicks needed to complete the main task: Logged in → Registered a Student → Course Enrolment → Captured Marks → Back to Dashboard.

Result: Main workflow requires 5 clicks from login to dashboard

Interpretation: Navigation was straightforward with clearly labelled buttons. No confusing menu structure was found.

Security Testing - Invalid Login Attempts Method:

Entered incorrect username/password 3 times on the login screen and observed the system response.

Result: System displayed "Invalid login credentials" popup for each invalid attempt. No account lockout occurred.

Interpretation: Basic validation was present, but the system does not lock the account after multiple failed attempts. This is a security risk for brute force attacks.

Identified Issue:

- The average is calculated incorrectly. When input is 100, 100, 100, on the 3 subject the result is 233.33334 instead of 100. This is because the code is adding the 3 marks and multiplying by 100, and not dividing by the number of subjects to get an average.
- Missing exception handling: `Convert.ToDouble()` will crash if the user enters non-numeric text like "a".

```

18
19 // Intentional weak validation and faulty average logic for testing purposes
20 MarkRecord record = new MarkRecord();
21
22 record.StudentId = txtMarkStudentId.Text;
23 record.StudentName = txtMarkStudentName.Text;
24 record.Subject1 = Convert.ToDouble(txtSubject1.Text);
25 record.Subject2 = Convert.ToDouble(txtSubject2.Text);
26 record.Subject3 = Convert.ToDouble(txtSubject3.Text);
27
28 // Intentional faulty calculation
29 record.Average = record.Subject1 + record.Subject2 + record.Subject3 / 3;
30
31 if (record.Average >= 50)
32 {
33     record.ResultStatus = "PASS";
34 }
35 else
36 {
37     record.ResultStatus = "FAIL";
38 }
39
40 txtMarksOutput.Text = "Marks processed successfully!";
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```

The Fix Explanation

- Brackets added: $(\text{subject1} + \text{subject2} + \text{subject3}) / 3$ ensures all three marks are added together before dividing by 3, giving the correct average.
- Input validation: `double.TryParse()` replaces `Convert.ToDouble()` so the app doesn't crash on invalid input. It shows a friendly error message instead.
- Exception handling: A try-catch block catches any unexpected errors and displays them to the user rather than crashing.

```

18 private void btnCalculateResults_Click(object sender, EventArgs e)
19 {
20     try
21     {
22         // Validate numeric input first
23         if (!double.TryParse(txtSubject1.Text, out double subject1) ||
24             !double.TryParse(txtSubject2.Text, out double subject2) ||
25             !double.TryParse(txtSubject3.Text, out double subject3))
26         {
27             MessageBox.Show("Please enter valid numeric marks for all subjects.");
28             return;
29         }
30
31         MarkRecord record = new MarkRecord();
32         record.StudentId = txtMarkStudentId.Text;
33         record.StudentName = txtMarkStudentName.Text;
34         record.Subject1 = subject1;
35         record.Subject2 = subject2;
36         record.Subject3 = subject3;
37
38         // Corrected calculation - add brackets for proper average
39         record.Average = (subject1 + subject2 + subject3) / 3;
40         record.ResultStatus = record.Average >= 50 ? "PASS" : "FAIL";
41
42         // This line must stay INSIDE the try block

```

```

43         txtMarksOutput.Text =
44             "Marks processed successfully!" + Environment.NewLine +
45             "Student ID: " + record.StudentId + Environment.NewLine +
46             "Student Name: " + record.StudentName + Environment.NewLine +
47             "Subject 1: " + record.Subject1 + Environment.NewLine +
48             "Subject 2: " + record.Subject2 + Environment.NewLine +
49             "Subject 3: " + record.Subject3 + Environment.NewLine +
50             "Average: " + record.Average.ToString("F2") + Environment.NewLine +
51             "Final Result: " + record.ResultStatus;
52     }
53     catch (Exception ex)
54     {
55         MessageBox.Show($"An unexpected error occurred: {ex.Message}");
56     }
57 }

```


ULMS Defect Report

Bug ID	Module	Description	Severity	Priority	Status	Resolution
BUG-01	Marks Capture	Average calculation is incorrect due to operator precedence. Input 100,100,100 returns 233.33 instead of 100	High	High	Fixed	Formula changed from Subject1 + Subject2 + Subject3 / 3 to (Subject1 + Subject2 + Subject3) / 3
BUG-02	Marks Capture	Application crashes with unhandled exception when non-numeric marks are entered. Error: "Input string 'a' was not in a correct format"	High	High	Fixed	Replace Convert.ToDouble() with double.TryParse() and add validation message
BUG-03	Login	System allows login with empty username and password fields and displays success message	Critical	High	Fixed	Add validation to check for empty fields before processing login and display error message
BUG-04	Course Enrolment	No account lockout after multiple invalid login attempts. System only shows "Invalid credentials" message repeatedly	Medium	Medium	Fail	Implement account lockout after 3 failed attempts or add CAPTCHA verification
BUG-05	Login	Variable scope error occurs when output text is placed outside try-catch block causing compilation error "record does not exist in current context"	Medium	Medium	Open	Move output assignment inside try block where record variable is in scope
BUG-06	Marks Capture	Empty marks field throws FormatException	Medium	Medium	Fixed	Add validation to check for empty/null marks fields before

						parsing. Use <code>double.TryParse()</code> and display error message "Please enter a valid mark" if field is empty or non-numeric.
--	--	--	--	--	--	---

Tools Used:

- Visual Studio 2022: IDE for running and debugging the ULMS application
- GitHub: Used to clone the project repository to the local machine
- Microsoft Word: Documentation and report writing
- Manual Testing: Test cases executed manually without automation tools

Corrections Implemented

- Added brackets to average formula: $(\text{subject1} + \text{subject2} + \text{subject3}) / 3$
- Replaced `Convert.ToDouble()` with `double.TryParse()` for input validation
- Added try-catch block for exception handling
- Added empty field validation on login
- Fixed variable scope by moving output assignment inside try block
- Added validation for empty marks fields in Marks Capture

Recommendations:

- Implement account lockout after 3 failed login attempts
- Add validation to prevent Course Enrolment with blank fields
- Run regression testing before production

Deployment Readiness Decision:

- Conditionally Ready for UAT
- Login, Marks Capture, and Reporting modules are ready for UAT.
- Course Enrolment validation and account lockout must be fixed before production.